

C-CLARKE INTERNATIONAL INSTITUTE OF DIGITAL SCIENCES

Deployment Guide

How to set up, run, and (eventually) deploy Smart Hire.

Program	Diploma in Advanced AI & Software Engineering
Batch	25/26
Project	Smart Hire — Service Marketplace
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1. Prerequisites

- Python 3.11+ and Node.js 18+
- A PostgreSQL database (the project uses a Supabase-hosted instance)
- An OpenAI API key (optional — the AI assistant degrades to a safe fallback message without one)

2. Backend Setup (Local Development)

1. Create and activate a virtual environment:

```
cd backend
python -m venv venv
source venv/bin/activate
```

2. Install dependencies:

```
pip install -r requirements.txt
```

3. Configure environment variables in `backend/.env` (see `.env.example`):

VARIABLE	PURPOSE
<code>DATABASE_URL</code>	PostgreSQL connection string
<code>JWT_SECRET_KEY</code>	Signs authentication tokens
<code>OPENAI_API_KEY</code>	Optional — enables the AI assistant

4. Start the API server:

```
uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

On startup, `Base.metadata.create_all()` creates any tables that don't yet exist (it does not alter existing tables — see the Database Scripts document for the full schema).

5. Verify: `curl http://127.0.0.1:8000/api/health` should return `200`.

3. Frontend Setup (Local Development)

```
1. cd frontend
   npm install
```

```
2. npx ng serve
```

3. Open `http://localhost:4200` . The dev server proxies API calls to `http://127.0.0.1:8000` .

4. Optional: MCP Server

The standalone MCP server exposes the same booking/search tools to external MCP clients (Claude Desktop, Claude Code):

```
cd backend
source venv/bin/activate
python mcp_server/server.py
```

5. Running the Test Suites

```
# Backend – 24 tests
cd backend && source venv/bin/activate && python -m pytest app/tests/ -v

# Frontend – 24 tests
cd frontend && npx ng test

# Smoke test against a running backend
python qa/smoke_test.py

# Real MCP protocol test
python qa/test_mcp_server.py
```

6. Production Deployment (Planned)

The application currently runs locally against a Supabase-hosted development database. Full production deployment is tracked as open work under the Version Control & Deployment epic on the project's Jira board, and would involve:

- Deploying the FastAPI backend to a Python-capable host (e.g. a container or PaaS supporting `uvicorn` /ASGI)
- Building the frontend for production (`ng build`) and serving the static output
- Pointing `DATABASE_URL` at a dedicated production Supabase project rather than the shared development one
- Configuring CORS, HTTPS, and environment secrets for the production hosts
- Running the full test suite and a smoke test against the deployed environment before go-live

This section is intentionally described as planned, not completed — the codebase and tests are production-ready, but the deploy step itself has not yet been executed.